Selective State-Space Models for TESS Exoplanet Vetting from Long Light Curves: A Controlled, Reproducible Feasibility Study

José Fabián Zumbado Ruiz and Jeremmy Aguilar Villanueva
School of Computing, Instituto Tecnológico de Costa Rica
Cartago, Costa Rica
{jzumbado, jaguilar}@estudiantec.cr

***Abstract*—Deep convolutional classifiers such as AstroNet and ExoMiner are central to automated vetting of transiting-planet candidates, but their effective receptive field is typically far shorter than a full TESS light curve (∼18,000 cadences per sector). Selective state-space models (Mamba) process such sequences in linear time with a global latent state, yet to our knowledge they have not been evaluated for confirmed-planet (CP) vs. false-positive (FP) vetting of TESS Objects of Interest (TOIs) from the *complete* global light curve. We present a controlled, reproducible feasibility study on 1,576 labelled TOIs under a single sealed, star-level (TIC) split. Against a stratified-prior baseline, a catalog-feature logistic regression, and an AstroNet-style single-branch CNN, a compact Mamba model (131K parameters) reaches a test AUC-ROC of 0.806 (95% bootstrap CI $[0.747, 0.857]$), versus 0.604 $[0.530, 0.678]$ for the CNN. The $+0.20$ gap is statistically significant (DeLong $p = 3.7 \times 10^{-8}$; paired-bootstrap ΔAUC CI $[+0.13, +0.28]$) and the two confidence intervals do not overlap. A larger AstroNet-style dual-branch baseline (with a phase-folded local view), trained without Kepler transfer learning, lands between the CNN and Mamba. Gradient-, integrated-gradient- and occlusion-based attributions concentrate on the transit dips for true positives. We explicitly frame this as a small-data feasibility comparison, not a state-of-the-art vetting catalog: the contribution is the first controlled evidence that selective state-space modeling of long TESS sequences yields a measurable, statistically significant advantage over tabular and shallow-convolutional baselines under realistic data and hardware constraints.**

***Index Terms*—exoplanet vetting, TESS, state-space models, Mamba, time-series classification, reproducibility**

I. Introduction

The Transiting Exoplanet Survey Satellite (TESS) observes >200,000 stars at 2-minute cadence, producing ∼18,000 photometric measurements per star per ∼27-day sector. Transit signals are shallow (∼0.01–1% flux dips) and morphologically degenerate with eclipsing binaries and instrumental systematics, which makes large-scale *vetting*—deciding whether an already-detected candidate is a confirmed planet (CP) or a false positive (FP)—a central bottleneck. Our task is vetting, not detection: NASA's pipeline performs the Box-Least-Squares search; we classify the resulting TESS Objects of Interest (TOIs) using their light curves together with the catalog metadata that exists *because* the TOI exists (orbital period, transit depth, epoch, stellar magnitude). Using catalog metadata as input is therefore not leakage in this setting, consistent with established practice [2], [3].

State-of-the-art vetting is built on 1D convolutional networks [2], [3], [5], [6]. A shallow single-branch CNN, however, sees the curve as a bag of local patches: stacking a few AstroNet-style blocks yields an effective receptive field of only ∼80 cadences (∼20 minutes of observation), far shorter than the multi-day spacing between transits in a sector (deeper or dilated stacks widen this, at additional cost). Transformers would model the full sequence but at $O(L^2)$ cost—prohibitive for $L = 18{,}000$ on commodity hardware. Selective state-space models (Mamba [1]) propagate an input-dependent latent state across the whole sequence in $O(L)$ time and constant memory, which is exactly the regime of long photometric series. To our knowledge, Mamba has not been evaluated for CP/FP vetting of TESS TOIs on the complete global light curve.

Contributions. (i) The first controlled comparison of a selective state-space model against tabular and convolutional baselines for TESS TOI vetting on full 18,000-cadence global curves, under one sealed star-level split. (ii) A statistically grounded evaluation—bootstrap confidence intervals for AUC-ROC/AUC-PR and DeLong tests for paired AUC differences—rather than point estimates alone. (iii) A close AstroNet-inspired dual-branch baseline adapted to the TESS setting as a strong upper-reference, plus a clearly labelled negative ablation (a naive Mamba+CNN late-fusion). (iv) Explainability (saliency, integrated gradients, occlusion) localizing the model's evidence to the transit dips. We deliberately make a *narrow, defensible* claim: under small data and a 4 GB GPU, Mamba extracts temporal signal that tabular features and a single-branch CNN miss; we do not claim a new vetting state of the art.

II. Related Work

Deep learning for transit vetting began with AstroNet [2], a dual-branch (global+local) CNN for Kepler that ranked candidates above classical features. ExoMiner [3] extended this with expert-diagnostic inputs and validated hundreds of new exoplanets; ExoMiner++ [5] ports the approach to TESS 2-minute data with Kepler-to-TESS transfer learning and additional diagnostics (periodograms, difference images, spacecraft attitude). For TESS specifically, Yu et al. [4] performed automated triage/vetting with CNNs, reporting an average precision of 69.3% in the vetting setting. Lighter, easier-to-replicate CNNs such as DART-Vetter [6] continue this line, and broader machine-learning validation frameworks have been applied to candidate validation [7]. These systems are strong and, in several cases, rely on large training sets and/or transfer learning; we therefore do not position our work as competing for state of the art.

Mamba [1] introduced selective state-space sequence modeling with linear-time scaling and a hardware-aware scan. It has been applied across language, audio and vision, but—to our knowledge—not to CP/FP vetting of TESS TOIs from the full global light curve. That gap is the niche this study addresses with a controlled, low-cost experimental comparison.

III. Data and Problem Setup

Sources. Labels come from the NASA Exoplanet Archive TOI catalog [12]; light curves are the PDCSAP_FLUX series retrieved from MAST [13] via `lightkurve` [11]. We define a binary task: confirmed planets (CP) as the positive class and false positives (FP) as the negative class, yielding 1,576 labelled TOIs after quality filtering (603 CP / 973 FP, a ∼1:1.6 imbalance).

Preprocessing. Each curve is normalized by its own median (no global dataset statistics, to avoid scale leakage across splits); NaNs and poor-quality cadences are masked/interpolated; all curves are fixed to $L = 18{,}000$ by truncation/padding. The *global view* is this full series. For the dual-branch ablations we additionally build a *local view* by phase-folding on the catalog (P, T_0), restricting to $|\phi - 0.5| \leq 2.5\,D/P$ and binning to 201 points; TOIs with missing/degenerate ephemerides or $P > 27.4\,\mathrm{d}$ (one sector) are excluded from the local-view subset.

Sealed, star-level split. A star may appear in several TESS sectors; splitting at the observation level would place the same TIC in train and test (severe leakage). We split at the *TIC* level, 70/15/15, preserving class balance, and we evaluate the test set *once* per model. The local-view subset is a strict subset of the same split (no TIC changes fold), so its test set ($N = 210$) is smaller than the global test set ($N = 237$); the two are not directly comparable and we report paired statistics only on shared samples. Table I gives the counts.

IV. Models

We evaluate a ladder of models in which each rung adds one design variable, so each AUC delta is attributable to a concrete component.

TABLE I
Star-level (TIC) splits. The local-view subset is a strict subset of the global split, filtered to TOIs with usable ephemerides ($P \leq 27.4$ d).

Split	Global N	CP/FP	Local-view N	CP/FP
train	1103	422/681	985	376/609
val	236	90/146	211	77/134
test (sealed)	237	91/146	210	78/132
total	1576	603/973	1406	531/875

Baselines. (1) *Stratified prior*: predicts the class prior; an empirical floor (AUC = 0.5). (2) *Catalog logistic regression*: ℓ_2-regularized logistic regression on *only* three catalog features (orbital period, transit depth, stellar magnitude), standardized with statistics fit on train only and class-balanced. Epoch (T_0) and transit duration (D) are *not* used by this baseline; they serve only to build the phase-folded local view. It answers whether catalog metadata alone separates CP from FP. (3) *CNN single-branch* (AstroNet-inspired, global view only): four `Conv1d` → BN → ReLU → MaxPool blocks (channels 16→32→64→128), global average pooling, and a small MLP head (62,881 parameters).

Mamba single-view (focus model). A linear embedding ($1 \to d_{\mathrm{model}}$=64), four residual Mamba blocks (d_{state}=16, d_{conv}=4, expand= 2) with LayerNorm, global average pooling, and a linear classifier (131,393 parameters). Gradient clipping (`max_norm`= 1.0) is essential: without it the model diverges to NaN on the long sequences. The state recursion $h_t = \bar{\mathbf{A}}_t h_{t-1} + \bar{\mathbf{B}}_t x_t, \;\; y_t = \mathbf{C}_t h_t$, with input-dependent $(\bar{\mathbf{A}}, \bar{\mathbf{B}}, \mathbf{C})$, lets the model integrate evidence across all 18,000 cadences and selectively down-weight noisy gaps.

Dual-branch ablations (local-view subset). (4) *AstroNet multibranch*: a close AstroNet-inspired dual-branch baseline following Shallue & Vanderburg [2], adapted to TESS—a deep global branch (five double-conv blocks, 16→256), a shallow local branch, concatenation, and a four-layer FC head (1,338,081 parameters). Because the original `Flatten` over $L = 18{,}000$ is infeasible, we use adaptive average pooling on the global branch; the model needs batch size 8, FP16 and gradient checkpointing to fit in 4 GB. (5) *ExoMamba V1 (negative ablation)*: a naive late-concat fusion of the Mamba global encoder with a small 3-block local CNN (∼153K parameters).

V. Experimental Protocol

All neural models are trained with class-weighted binary cross-entropy and AdamW, with early stopping on validation AUC-ROC; the test set is untouched during development. We do not use k-fold cross-validation: a single Mamba run takes ∼1 h on an RTX 3050 (4 GB), so k=5 over a hyperparameter sweep would cost ∼100 GPU-hours. As a statistical proxy we report *multi-seed* variability—five seeds for Mamba single-view, three for the dual-branch ablations—and a probability-averaged *ensemble* per model family. The Mamba hyperparameters were chosen by a small structured search on validation (learning rate 1.5×10^{-3}; d_{state}=16; patience 10), not an automated tuner, again for cost.

Inference statistics. On the sealed test set we report 95% bootstrap confidence intervals (2,000 stratified resamples) for AUC-ROC in Table II; AUC-PR is listed as a point estimate, and its bootstrap CI is computed identically and included in the released statistics file. For paired model comparisons on shared samples we report DeLong's test [9] (fast formulation [10]) for the AUC difference, together with a stratified paired-bootstrap CI for ΔAUC.

VI. Results

A. Main comparison

Table II reports test metrics with bootstrap CIs. Mamba single-view reaches AUC-ROC 0.806 (ensemble) and 0.810 (best seed), with a confidence interval that lies entirely above the CNN's confidence interval and above the logistic-regression point estimate. The CNN barely matches catalog logistic regression (0.604 vs. 0.605): on this dataset, a shallow global CNN adds essentially nothing over three catalog numbers, whereas the state-space model does.

B. Statistical significance

Table III reports paired comparisons on shared samples. The headline result is unambiguous: Mamba ensemble vs. CNN single-branch gives $\Delta\mathrm{AUC} = +0.20$ with DeLong $p = 3.7\times10^{-8}$ and a paired-bootstrap CI of $[+0.13, +0.28]$ that excludes zero by a wide margin. Two secondary findings are also significant: the local view helps the CNN family (AstroNet vs. CNN, +0.11, $p = 0.009$), and Mamba on the shared 210-sample subset still beats the 1.34M-parameter AstroNet (+0.09, $p = 0.011$). Conversely, the ensemble does *not* significantly outperform the best single seed ($\Delta = +0.04$, $p = 0.048$, CI touching zero): ensembling mainly reduces seed variance—it averages out a single weak seed (AUC 0.628)—rather than raising the ceiling. We report this honestly rather than crediting the ensemble with the gain.

C. Calibration, confusion and error structure

The Mamba ensemble is also better calibrated (Brier 0.192 vs. 0.271 for the CNN; see Fig. 2). Its improvement is qualitative, not just ranking: relative to the CNN it roughly halves false positives (55 vs. 96) and nearly doubles true negatives (91 vs. 50) on the same 237 test stars, i.e. it rejects catalog FPs the CNN accepts. An automated error breakdown (Fig. 3) localizes the dominant failure mode: transits with depth $\in (136, 770]$ ppm have a 45.8% error rate ($n = 48$)—shallow transits near the TESS photometric noise floor. The hardest false positives are eclipsing binaries with deep, symmetric dips, the known information limit of single-view vetting.

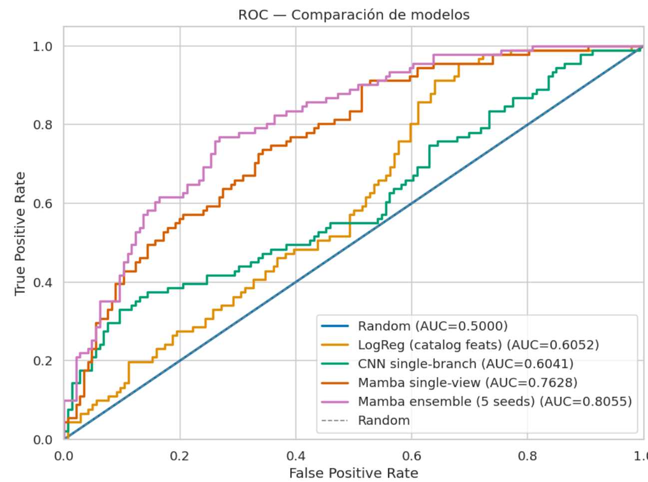


Fig. 1. Test ROC curves for the global-view models. Mamba (upper-left) is clearly separated from the CNN and logistic regression (near the diagonal); the visual gap is the +0.20 AUC difference quantified in Table III.

D. Explainability

Applying gradient saliency, integrated gradients [8] and occlusion sensitivity to the best Mamba seed over eight confident test cases (top-2 per TP/TN/FN/FP quadrant), the three methods agree (Fig. 4): for true positives the attribution concentrates on the periodic transit dips—the model "looks in the right place"; for false negatives it is weak and diffuse; for false positives it correctly fires on real dips but cannot separate planetary transits from eclipsing binaries, consistent with the error analysis.

E. Negative ablation: naive Mamba+ CNN late fusion

ExoMamba V1 (Table II, last row) collapses *below chance* (0.460 $[0.382, 0.541]$) across three seeds, despite passing a sanity overfit (val AUC = 1.0 on 64 examples), ruling out an implementation bug. We read this as the fusion operator, not the branches, being at fault: a tiny concat-MLP head lets the fast-training local CNN dominate the gradient and starve the Mamba encoder. The dual-branch AstroNet, with a ∼700K-parameter FC head, instead recovers the local-view signal (+0.11 over the single-branch CNN). The lesson is that combining selective state-space and convolutional features needs a non-trivial, jointly trained fusion—naive late concatenation is the wrong operator—which motivates gated/attention fusion as future work rather than a headline result.

VII. Discussion

The +0.20 AUC gap over the CNN is not explained by capacity (Mamba 131K vs. CNN 63K, only 2×). It is architectural: our single-branch CNN's effective receptive field (∼80 cadences) cannot see the multi-day periodicity of transits, while Mamba's $O(L)$ state recursion integrates all transits in a sector into one latent trajectory. A much

TABLE II

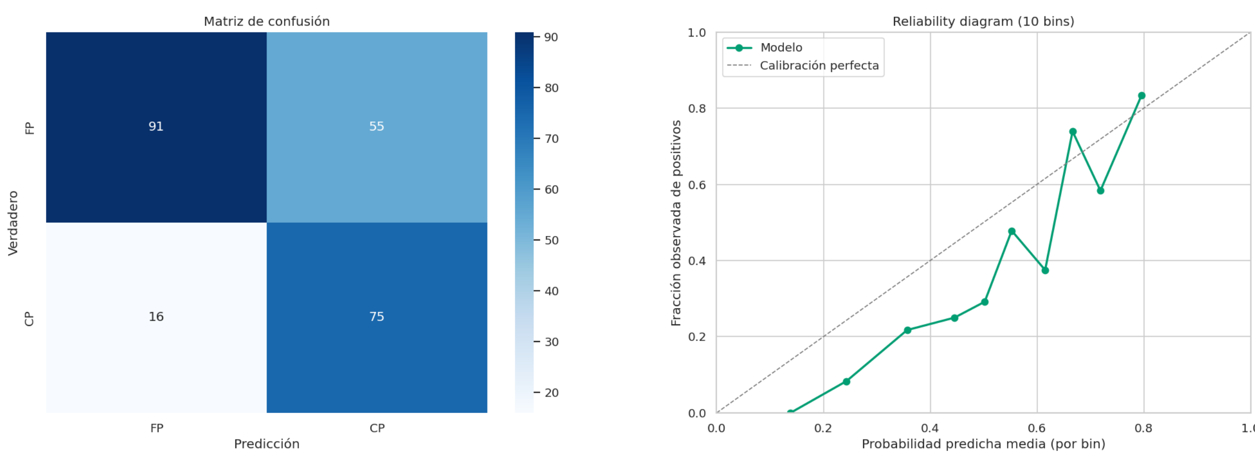
Sealed test results. Global-view models use N=237; dual-branch models use the strict local-view subset N=210 (not directly comparable; see paired tests in Table III). †Logistic-regression predictions were stored as a summary without per-sample probabilities, so no bootstrap CI is computed. *Negative ablation (below chance); recall = 1.0 reflects a degenerate all-positive collapse.

Model	View	N	AUC-ROC [95% CI]	AUC-PR	F_1	Recall	Brier
Stratified prior	–	237	0.500 [0.500, 0.500]	0.384	0.000	0.000	0.237
Catalog logistic regression	catalog	237	0.605 †	0.464	0.486	0.571	–
CNN single-branch	global	237	0.604 [0.530, 0.678]	0.551	0.539	0.758	0.271
Mamba single (single seed)	global	237	0.763 [0.697, 0.826]	0.650	0.633	0.835	0.210
Mamba single (5-seed mean±std)	global	237	0.750 ± 0.065	–	–	–	–
Mamba single (ensemble, 5)	global	237	**0.806 [0.747, 0.857]**	**0.711**	**0.679**	0.824	**0.192**
Mamba single (best seed)	global	237	0.810 [0.756, 0.862]	0.722	0.661	0.802	0.190
AstroNet multibranch (ensemble, 3)	global+local	210	0.716 [0.644, 0.779]	0.582	0.563	0.603	0.217
ExoMamba V1 (ensemble, 3) *	global+local	210	0.460 [0.382, 0.541]	0.371	0.542	1.000	0.254

TABLE III

Paired AUC-ROC comparisons on shared test samples. DeLong's test with stratified paired-bootstrap confidence intervals on ΔAUC.

Comparison (A vs. B)	N	ΔAUC [95% CI]	DeLong p
Mamba ens. vs. CNN	237	+0.20 [+0.13, +0.28]	3.7×10^{-8}
Mamba best vs. CNN	237	+0.21 [+0.13, +0.29]	1.5×10^{-7}
Mamba ens. vs. AstroNet	210	+0.09 [+0.02, +0.16]	0.011
AstroNet vs. CNN	210	+0.11 [+0.03, +0.19]	0.009
AstroNet vs. ExoMamba V1	210	+0.26 [+0.16, +0.36]	9.2×10^{-7}
Mamba ens. vs. best seed	237	+0.04 [−0.00, +0.08]	0.048



(a) Confusion matrix

(b) Reliability diagram

Fig. 2. Mamba ensemble on the sealed test set.

larger dual-branch CNN (AstroNet, 1.34M) with an explicit local view narrows but does not close the gap, and—without Kepler-to-TESS transfer learning, which the strongest published systems rely on—underfits this small training set. Taken together, on ∼1.6K labelled TOIs and a 4 GB GPU, long-range sequence modeling of the complete global curve is the most effective single design choice we tested.

We stress what this does *not* show. It is not a vetting state of the art: AstroNet/ExoMiner-class systems use far larger datasets, transfer learning and richer diagnostics. It is a controlled feasibility result: under matched data, a single sealed split and quantified uncertainty, selective state-space modeling beats tabular and shallow-convolutional baselines by a margin that survives DeLong and bootstrap testing.

VIII. Limitations

Small data. 1,576 labelled TOIs (∼1/10 of Kepler-scale training sets) is the main constraint; absolute AUC (0.806) is below what large-data, transfer-learning systems report, as expected. **Label selection bias.** We use CP vs. FP;

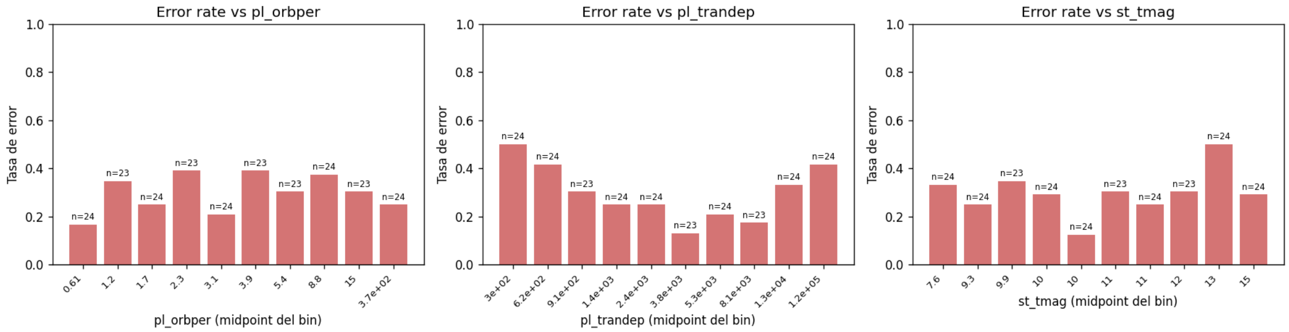


Fig. 3. Error rate vs. binned period, transit depth and stellar magnitude. Errors concentrate in shallow transits near the noise floor.

Fig. 4. Attribution summary (8 cases × 3 methods) for the best Mamba seed. Red/blue indicate evidence toward "planet"/"not planet". True positives place mass on the transit dips.

confirmed planets tend to be better-studied and cleaner than the broader, more operationally relevant PC-vs-FP (candidate triage) problem, so our labels likely make the task somewhat easier and less representative of live triage. **No transfer learning** from Kepler (no reliable public checkpoint), which disadvantages the dual-branch CNN in particular. **No external validation** on held-out sectors or an independent catalog, and **no common benchmark** against ExoMiner++/DART-Vetter under identical inputs. **No *k*-fold CV** (compute-bound), mitigated by multi-seed reporting. These bound the claim to a within-study, same-split comparison.

IX. Conclusion and Future Work

Under a sealed star-level split, quantified uncertainty and matched data, a compact selective state-space model is a computationally feasible sequence-modeling alternative for long TESS light curves, significantly outperforming tabular and shallow-convolutional baselines for CP/FP TOI vetting. Future work: a non-naive Mamba+CNN+scalar fusion with a mandatory scalar-only control to detect shortcut learning; Kepler-to-TESS transfer; extension to the PC-vs-FP triage setting; and external, cross-sector validation toward a usable follow-up list.

Reproducibility. All splits are versioned at the TIC level; every run stores its config, git commit, environment, training curve, checkpoints and per-sample test predictions; the test set is loaded once behind an explicit guard; 49 unit tests cover the data and metrics paths. Inference statistics are computed from the stored predictions by a single script, so all confidence intervals and DeLong tests in this paper are reproducible without retraining.

References

[1] A. Gu and T. Dao, "Mamba: Linear-Time Sequence Modeling with Selective State Spaces," arXiv:2312.00752, 2023.

[2] C. J. Shallue and A. Vanderburg, "Identifying Exoplanets with Deep Learning: A Five-planet Resonant Chain around Kepler-80," *Astron. J.*, vol. 155, no. 2, p. 94, 2018.

[3] H. Valizadegan *et al.*, "ExoMiner: A Highly Accurate and Explainable Deep Learning Classifier to Mine Exoplanets," *Astrophys. J.*, vol. 926, no. 2, p. 120, 2022.

[4] L. Yu *et al.*, "Identifying Exoplanets with Deep Learning. III. Automated Triage and Vetting of TESS Candidates," *Astron. J.*, vol. 158, no. 1, p. 25, 2019.

[5] H. Valizadegan *et al.*, "ExoMiner++ on TESS with Transfer Learning from Kepler: Transit Classification and Vetting Catalog for 2-min Data," arXiv:2502.09790, 2025.

[6] S. Fiscale *et al.*, "DART-Vetter: A Deep LeARning Tool for automatic triage of exoplanet candidates," arXiv:2506.05556, 2025.

[7] D. J. Armstrong, J. Gamper, and T. Damoulas, "Exoplanet Validation with Machine Learning: 50 new validated Kepler planets," arXiv:2008.10516, 2020.

[8] M. Sundararajan, A. Taly, and Q. Yan, "Axiomatic Attribution for Deep Networks," in *Proc. Int. Conf. Mach. Learn. (ICML)*, 2017.

[9] E. R. DeLong, D. M. DeLong, and D. L. Clarke-Pearson, "Comparing the Areas under Two or More Correlated Receiver Operating Characteristic Curves," *Biometrics*, vol. 44, no. 3, pp. 837–845, 1988.

[10] X. Sun and W. Xu, "Fast Implementation of DeLong's Algorithm for Comparing the Areas Under Correlated Receiver Operating Characteristic Curves," *IEEE Signal Process. Lett.*, vol. 21, no. 11, pp. 1389–1393, 2014.

[11] Lightkurve Collaboration, "Lightkurve: Kepler and TESS time series analysis in Python," Astrophys. Source Code Libr., ascl:1812.013, 2018.

[12] NASA Exoplanet Archive, "TESS Mission (TOI Catalog)." [Online]. Available: https://exoplanetarchive.ipac.caltech.edu/docs/TESSMission.html

[13] MAST Archive, "TESS Mission Data Archive." [Online]. Available: https://archive.stsci.edu/missions-and-data/tess